Stage 1: Quality-Aware Preprocessing

Raw Multi-Sequence MRI

110 Patients | 3,929 Slices
Pre-contrast, FLAIR, Post-contrast
Resized: $128 \times 128 \times 3$

Morphological Artifact Screen

Canny Edge Density ($ED < 0.003$)
Contour Ratio ($CR < 0.01$)
Centroid Dist ($d_c > 0.30$)
Flags 200 noisy slices (199 tumor-free)

Quality-Aware Wavelet Denoising

$Q(I) = \frac{1}{2}[\sigma(I - \mathcal{M}_{3\times3}(I)) + \sigma(I - \mathcal{G}_{3\times3}(I))]$
If $Q(I) < 4.0$: BayesShrink 2-level (95.1%)
If $Q(I) \geq 4.0$: High-contrast bypass (4.9%)

Stage 2: Attention Residual U-Net Backbone (8.20M Params)

ResBlock 1 (128^2)
32 Filters | DropPath: 0.00
GroupNorm ($G = 8$) + GELU

MaxPool 2x2

ResBlock 2 (64^2)
64 Filters | DropPath: 0.05
GroupNorm ($G = 8$) + GELU

MaxPool 2x2

ResBlock 3 (32^2)
128 Filters | DropPath: 0.10
GroupNorm ($G = 8$) + GELU

MaxPool 2x2

ResBlock 4 (16^2)
256 Filters | DropPath: 0.15
GroupNorm ($G = 8$) + GELU

MaxPool 2x2

Bottleneck (8^2)
512 Filters | DropPath: 0.20
Spatial Dropout ($p = 0.25$)

UpConv 2x2

Decoder 4 (16^2)
256 Filters | ConvTrans 2x2
ResBlock + GN ($G = 8$)

UpConv 2x2

Decoder 3 (32^2)
128 Filters | ConvTrans 2x2
ResBlock + GN ($G = 8$)

UpConv 2x2

Decoder 2 (64^2)
64 Filters | ConvTrans 2x2
ResBlock + GN ($G = 8$)

UpConv 2x2

Decoder 1 (128^2)
32 Filters | ConvTrans 2x2
ResBlock + GN ($G = 8$)

AG1 | AG2 | AG3 | AG4

Deep Supervision Heads

$\mathcal{L}_{total} = 1.0\mathcal{L}_{main} + 0.3\mathcal{L}_{d2} + 0.3\mathcal{L}_{d3}$
$\mathcal{L} = 0.70\mathcal{L}_{FTL}(\alpha = 0.8, \beta = 0.2) + 0.30\mathcal{L}_{BF}$

Logits $z(x)$

Stage 3: Calibration & Anisotropic 3D Hysteresis

Nested Temperature Scaling (T^*)

Fitted on inner validation logits ($T^* = 1.844$)
$\hat{P}(x) = \sigma(z(x)/T^*)$
Reduces ECE by 46.9% (0.369% → 0.196%)
Zero impact on ROC / test labels

1. In-Plane Binarization (XY-Plane)

$M^{(0)}(x, y, z) = \mathbb{I}(\hat{P} \geq \tau^*_{xy,k})$
Fold-tuned $\tau^*_{xy} = 0.74$ (Inner Val)
Shaves overconfident False-Positive halo
Boosts Precision: 71.95% → 76.86%

2. Anisotropic Axial Boundary Rescue

If slice z empty, check adjacent $z \pm 1$:
$S_{prior} = \text{Dilate}_{2D}(M^{(0)}_{z-1} \cup M^{(0)}_{z+1}, \mathcal{K}_{3\times3})$
Rescue: $M^{(1)}_z = M^{(0)}_z \cup (S_{prior} \cap \mathbb{I}(\hat{P}_z \geq \tau^*_{z,\text{low}}))$
Decouples in-plane $\tau^*_{xy} = 0.74$ from axial $\tau^*_{z,\text{low}} = 0.35$

3. 3D Component Filtering & Fallback

26-connected 3D component analysis:
Keep C_l iff $|C_l| \geq V^*_{min,k}$ AND $\max\hat{P} \geq \tau^*_{seed,k}$
Fallback: Retain $\max\hat{P} \geq \tau^*_{seed}$ if volume empty
Reduces empty-slice FPR by 42.4% (13.65% → 7.86%)

Volumetric 3D Tumor Mask

$\text{Dice}_{pos} = 74.92\%$, $\text{Dice}_{all} = 86.12\%$, 3D Dice = 77.73%
Clinical ICC = 0.8894, Genomic $p = 0.0268$